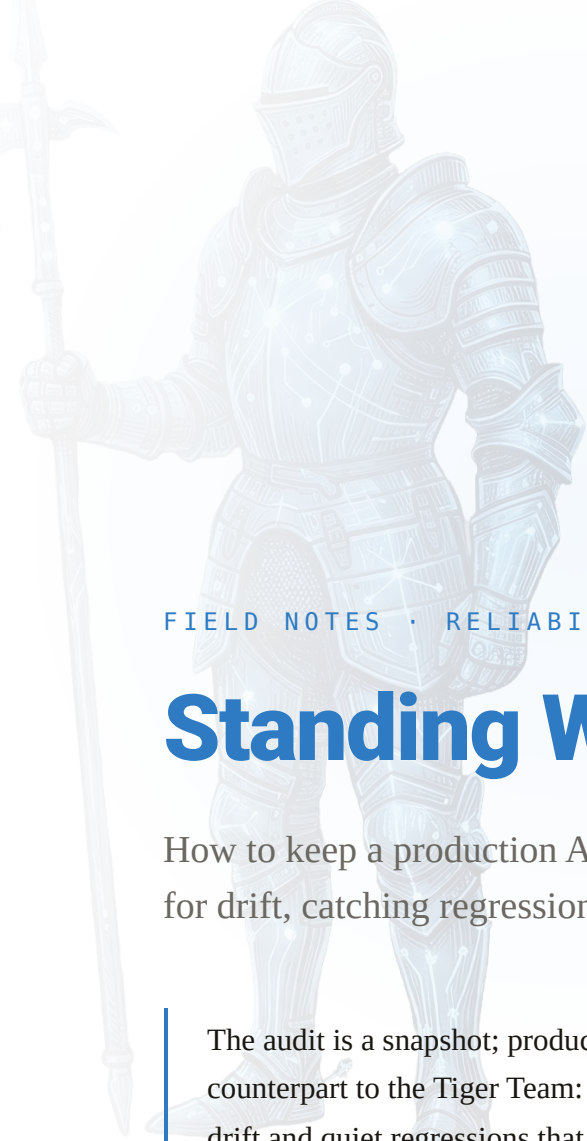




FIELD NOTES • RELIABILITY & DEFENSE

Standing Watch

How to keep a production AI system honest between audits — watching for drift, catching regressions early, and shipping fixes that actually hold.

The audit is a snapshot; production is a movie. This paper documents the defensive counterpart to the Tiger Team: a monitoring layer that never leaves its post, catches the drift and quiet regressions that creep in between scheduled hunts, and is rewarded for the failures that never reached a customer.

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Series • [Who Audits the Robots?](#)

Subject • Continuous monitoring for multi-agent AI systems

Companion • “Paying Agents to Find Truth”

A *sentinel* is a guard who stands and keeps watch — posted at a boundary to sense danger before it gets in. The word traces to the Italian *sentire*, “to perceive.” A sentinel doesn’t roam or hunt. It holds a position, and it never leaves it.

That is the exact posture of this layer. The Tiger hunts on a schedule; the Sentinel watches every hour in between. It doesn’t attack the system — it guards it, catching the drift and the quiet regressions that creep in after the audit truck has driven off. The story, again, is one an executive can repeat: *we keep a guard on the wall so nothing slips in between inspections*. The engineering underneath is how you keep that guard honest, too.

“The audit is a **snapshot**; production is a movie.”

THE PROBLEM, IN ONE LINE

01 Passing once is not staying honest

A clean audit is a moment, not a guarantee.

A Tiger Team audit tells you the system was sound on the day it was inspected. But production systems don’t hold still. Models get updated underneath you. Traffic shifts toward inputs no one tested. A prompt tweak that fixed one flow quietly breaks another. None of this announces itself — the system keeps answering in the same confident tone it always had.

This slow, invisible decay is *drift*, and it is the reason a single passing grade expires. The dangerous failures are rarely loud crashes; they are the regressions that look fine until a customer hits the one path that no longer works. The Sentinel exists to close the gap between two audits — to make sure “it passed last month” still means something today.

02 What the Sentinel watches for

The Sentinel doesn't re-run the full hunt every hour — that's the Tiger's job. It watches a small set of leading signals, the ones that move first when something is going wrong.

- ▶ **Drift.** The system's behavior slides away from its audited baseline — answers get longer, slower, or subtly off — without any code change to blame.
- ▶ **Regressions.** A fix or update repairs one path and silently breaks another. The Sentinel re-checks the paths a recent change touched, not the whole surface.
- ▶ **Silent failures.** The most dangerous class: the system returns a confident answer that is simply wrong, with no error thrown. These never show up in an uptime dashboard.
- ▶ **Boundary events.** Anything crossing the line into production — a new model version, a prompt edit, a schema change — trips a watch before it's trusted.

“Drift doesn't announce itself. The system keeps answering in the same confident tone it always had.”

DESIGN PRINCIPLE · SENTINEL

WATCHING LEADING SIGNALS, NOT LAGGING ONES

An uptime monitor tells you the system is *up*. It cannot tell you the system is *right*. The Sentinel watches correctness signals — baseline drift, confirmed regressions, silent wrong answers — because those move long before a customer complaint or a revenue dip ever shows up in a report.

03 The gate: from finding to fix

Catching a problem is only half the job. The Sentinel's real value is what happens next — a disciplined path from “something looks wrong” to “it's fixed, and the fix held.” Nothing skips the gate.

```
promote(change) if reviewed && staged_ok && rollback_ready
```

reviewed — a human signs off on the diagnosis and the proposed fix

staged_ok — the fix proves out on a slice of traffic before full rollout

rollback_ready — a tested one-step revert exists before anything ships

Each condition closes a way a “fix” can make things worse:

- ▶ **Human review** — no autonomous patching of a live revenue path. The Sentinel proposes; a person approves.
- ▶ **Staged rollout** — a fix earns full traffic by surviving a small slice first, so a bad repair is contained, not global.
- ▶ **Rollback ready** — the revert is written and tested before the change ships, so recovery is one step, not an incident.

“A watchtower is only useful if someone's [in it](#) — and if the alarm comes with a way down.”

ON THE GATE

WHY ROLLBACK COMES FIRST

The instinct is to ship the fix and write the rollback later. That order is backwards. A revert authored *after* a bad deploy is written in a panic; one authored *before* is written calmly and tested. The gate refuses to promote a change until the way back already exists.

04 The opposite incentive

The Tiger and the Sentinel are both paid to win — but winning means opposite things, and that is what keeps each one honest. The Tiger wins by finding a real, reproducible hole. The Sentinel wins the other way: by the regressions that never shipped and the drift caught before a customer ever felt it.

“Reward what **didn’t break, not just what got fixed.”**

DESIGN PRINCIPLE · SENTINEL

This matters because the naive incentive — rewarding the Sentinel for fixes shipped — recreates the Tiger’s original problem in reverse: a guard paid per repair has a reason to let small fires start so it can be seen putting them out. So the Sentinel is scored on a *quiet watch*: uptime of correctness, mean time to catch, and regressions stopped at the gate. A boring month is a winning month.

TWO WINNERS, OPPOSITE VICTORIES

Both sides are paid to win — but the Tiger wins by breaking things and the Sentinel wins by nothing breaking. Neither can cheat, because the other is watching. That opposition, not any single score, is what makes the whole system hard to game.

05 What it produces

Between Tiger audits, the Sentinel resolves to three standing artifacts:

- ▶ **A live health signal** — a continuously updated read on whether production still matches its audited baseline, not a monthly snapshot.
- ▶ **A change log with receipts** — every promoted fix, what it touched, who approved it, and the evidence it held after rollout.
- ▶ **A tested way back** — a working, current rollback for every live change, so recovery is never improvised.

Put the two papers together and the picture is complete: the Tiger proves the system was sound when it was inspected; the Sentinel proves it has *stayed* sound since. One hunts, one guards. Between them, “we think it’s working” becomes “we know it is — and we’d catch it the hour it wasn’t.”

The Tiger hunts on a schedule. The Sentinel never sleeps. — companion paper: “Paying Agents to Find Truth.”